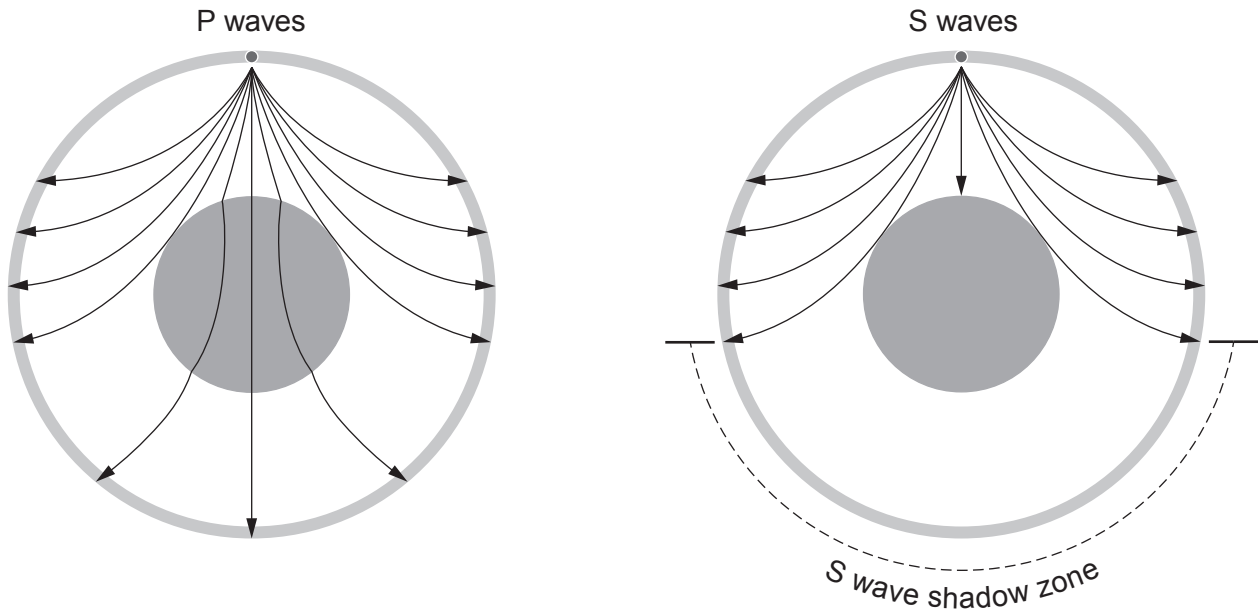


2. The diagram shows the path of seismic P and S waves through the Earth.



(a) (i) Underline the correct word in the bracket.

[1]

The Earth has a liquid (**core** / **mantle** / **crust**).

(ii) Tick (✓) the **three** correct statements about P and S waves.

[3]

P waves are transverse waves.

☐

P waves travel faster than S waves.

☐

S waves cannot travel in a liquid.

☐

P waves cannot travel in a solid.

☐

S waves are transverse waves.

☐

P waves cannot travel in a liquid.

☐


- (b) (i) P waves from an earthquake, which are travelling at a mean speed of 8000 m/s, reach a seismic monitoring station 500 s later.

Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the P waves to the monitoring station. [2]

distance = m

- (ii) **A**, **B** and **C** are monitoring stations in Wales.
They collect data from an earthquake.

The epicentre is located somewhere on the circles shown in the diagram below.
Mark with a cross (x) the location of the epicentre. [1]

